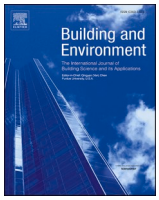
Building and Environment 222 (2022) 109385



Contents lists available at ScienceDirect

Building and Environment

journal homepage: www.elsevier.com/locate/buildenv





A novel methodology for human thermal comfort decoding via physiological signals measurement and analysis

Silvia Angela Mansi [a], Ilaria Pigliautile [b,c], Marco Arnesano [a], Anna Laura Pisello [b,c,d,*]

[a] Università Telematica eCampus, Novedrate, CO, Italy
[b] CIRIAF, Interuniversity Research Center on Pollution and Environment "Mauro Felli", University of Perugia, Perugia, PG, Italy
[c] Department of Engineering Department, University of Perugia, Italy
[d] Department of Civil and Environmental Engineering, Princeton University, Princeton, NJ, United States

ARTICLE INFO

Keywords:
Thermal comfort
Physiological measurements
Wearables
EEG
HRV
EDA
Thermal sensation
Energy efficiency in buildings
Personal comfort model

ABSTRACT

Personal comfort models (PCM) represent the most promising paradigm for human-centric thermal comfort in buildings. Several data sources can be used to generate a PCM: environmental data, physiological data, occupants' response. Advances in wearable sensing suggest that the use of physiological data for real time comfort measurement can be the start-up of the next generation of building design and operation with PCMs. However, proof of evidence about the adoption of non-invasive but accurate measurement methods and about correlations between physiological features and thermal sensation, are still required. This study presents the results from a large original experimental campaign aiming at human thermal comfort decoding via physiological signal. Two non-invasive wearables were used to simultaneously measure four key physiological signals (electroencephalography (EEG), Heart Rate Variability (HRV), electrodermal activity (EDA) and skin temperature (ST) on 52 subjects exposed to three different thermal conditions (namely cold, warm, and neutral) in a controlled environment. Data acquired from 219 tests were therefore analysed to determine the statistical importance of physiological features. Results showed that cold and warm thermal sensations can be uniquely identified by each physiological signal; while neutral sensation is the less distinguishable. More specifically, statistical differences (p-value <0.01) between cold and warm conditions were detected for the first time among EEGs features (Beta TP10, Gamma TP10 relative alpha TP9), time- and frequency-domain features of HRV, EDA tonic component and mean ST. Experimental results finally demonstrated that physiological measurements can identify specific thermal sensation, of crucial importance for the most advanced PCMs and for disclosing novel energy saving opportunities, accounting for people's diversities.

1. Introduction

Human beings are continuously exposed to a variety of environmental stimuli, which affect their health condition and potential behaviour [1], including personal and social interactions [2] with their surroundings and their capability of coping with specific social [3], personal, or working tasks [4] influencing building energy demand and not only [5]. The strong bonds between human health status and environmental stressors are already recognized in different domains [6]. Recent findings in building physics highlight the role of human environmental perception as a key driver in occupants' behaviour and interaction with building systems [7,8]. Introducing the human dimension in the building management system is crucial in determining actual building operation [9] and to minimize the energy consumption of building systems [10,11], while maximizing the Indoor Environmental Quality (IEQ) and occupants' comfort perception, satisfaction [12,13], and well-being [14,15]. To comprehensively address the individual perception of the surroundings, it is mandatory to explore in-depth the mechanism behind human environmental perception accounting for simultaneous occurrences of multiple stimuli belonging to different domains of comfort [16], and the human reaction to these stimuli resulting from individual psychological and physiological responses. All these individual differences lead to diversity in environmental perception as highlighted in the review by Schweiker at al. specifically focusing on thermal perception diversities [17]. All this considered, it is almost impossible to find out a model capable of interpreting the needs and preferences of all the occupants, especially in

* Corresponding author. CIRIAF, Interuniversity Research Center on Pollution and Environment "Mauro Felli", University of Perugia, Perugia, PG, Italy.
E-mail address: anna.pisello@unipg.it (A.L. Pisello).

https://doi.org/10.1016/j.buildenv.2022.109385
Received 21 April 2022; Received in revised form 7 June 2022; Accepted 7 July 2022
Available online 11 July 2022
0360-1323/© 2022 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Abbreviations

Ag	silver
ANS	Autonomic Nervous System
BLE	Bluetooth Low Energy
BP	Blood Pressure
BVP	Blood Volume Pulse
CNS	Central Nervous System
CO_2	Carbon Dioxide
DSP	Digital Signal Processing
ECG	Electrocardiogram
EDA	Electrodermal activity
EEG	Electroencephalography
ESC	European Society of Cardiology
FFT	Fast Fourier Transform
GSR	Galvanic Skin Response
GUM	Guide to the expression of uncertainty in measurement
HF	High Frequency Power
HRV	Heart Rate Variability
HR	Heart Rate
HVAC	Heating Ventilation and Air Conditioning
IBI	Inter Beat Interval
IEQ	Indoor Environmental Quality
LED	Light Emitting Diode
LF	Low Frequency Power
LF/HF	Low to High frequency ratio
NASPE	North American Society of Pacing and Electrophysiology
PCM	Personal Comfort Model
PNS	Parasympathetic Nervous System
PPG	Photoplethysmography
PSD	Power Spectral Density
RGB	Red-Green-Blue
SC	Skin Conductance
SCL	Skin Conductance Level
SCR	Skin Conductance Response
SNS	Sympathetic Nervous System
ST	Skin Temperature
TS	Thermal Sensation
VLF	Very Low Frequency Power

high-quality new constructions where minor environmental stresses are expected [18].

Personal Comfort Models (PCMs) have the potential of solving this issue. As reported by Joyce et al. [19], PCMs rely upon the individual as the unit of the analysis, taking advantage directly of the individual's feedback (sensation, preference, and acceptability), physiological signals records [20], or personal comfort systems control behaviour [21] and relevant information (mainly personal, environmental, and building systems technical specifics) to train a data-driven model. Nevertheless, still most of the PCM analyses are performed by taking into account environmental-only variables as measured individually or perceived.

At the same time, the recent development of non-invasive wearable sensing technologies highlights the potential of monitoring specific human physiological signals throughout daily life [22], having, therefore, more reliable signals which can be correlated to human preference and perception, rather than potentially biased surveys-only data sources. Electroencephalography (EEG), Heart Rate Variability (HRV), Skin Temperature (ST), and Electrodermal Activity (EDA) have been identified as the main responsiveness physiological signals reflecting the thermoregulatory system activity [22].

EEG is a non-invasive method to record the electrical potential difference between electrodes attached to a subject's scalp [23] according to the International 10–20 system [24], (range of 10 μV–100 μV of amplitude) resulting in the so-called "brainwaves". EEGs can be analysed through delta, theta, alpha, beta, and gamma waves, each of these brain waves corresponds to different states of thought or experience [25]. Experimental studies on brainwaves and human thermal perception predominately investigated the Power Spectrum Densities (PSDs) variation related to the different ambient temperature exposure of participants. In general, results revealed that different environmental temperatures correspond to an increase or decrease of power in particular brain frequency bands [26–28]. Son and colleagues [29] showed a significant increase in the relative theta when the subject felt pleasure thermal sensation. Other studies revealed that participants showed higher values of alpha power in pleasant thermal environments, while beta power has lower values [30,31].

Conventionally, electrocardiogram (ECG) is the method used to record the heart's electrical activity, from which the Heart Rate (HR) and HRV can be derived [32]. HRV represents the physiological phenomenon of the variation, in the time interval, between consecutive heartbeats in milliseconds. With the advent of wearable devices, photoplethysmography (PPG) has recently gained interest in HR-HRV measurements. The agreement between PPG and ECG-based HR-HRV has been demonstrated in several studies [33,34]. When attached to the skin, the PPG sensor measures the change in blood volume caused by the heart's pumping by transmitting infrared light signals and measuring the absorption of the reflected light by the blood [35]. This method allows the calculation of the rate of blood flow as controlled by the heartbeats, also called Blood Volume Pulse (BVP), and the detection of a peak in that pulse. The interval distance between those peaks is known as Inter Beat Interval (IBI) which defines the HR. Several studies confirmed the rule of HRV to guide the heart's adaptation to physiological and environmental events, (e.g., breathing, exercise, physical and mental stress, and metabolic alteration) [36] and the Autonomic Nervous System (ANS) activity Experimental studies on HRV and human thermal perception have mainly focused on how time and frequency domain features vary as participants were exposed to different environmental thermal conditions. The results from Yao et al. [37] and Weiwei et al. [38] study showed higher values of low to high-frequency ratio (LF/HF) in extreme (hot and cold) thermal environments and lower values in the neutral condition. Other studies [39,40] revealed value of LF/HF near revealed a value of LF/HF near 1 in comfortable thermal conditions and a higher value when subject perceived thermal discomfort.

ST and EDA can be defined as indicators of the thermal homeostasis actions occurring at the skin surface. Measuring both of those signals, by applying a proper sensor directly on the skin surface, can contribute to investigating some processes linked to the thermoregulation system activities. While ST represents the pure measure of the temperature of the skin, EDA is the change in sweat gland activity [41]. EDA reflects the intensity of the human emotional state; stimulation of the sweat gland can in turn increase the electrical conductivity of the skin. In the field of thermal comfort investigation using skin physiological signals, it is indicated that both ST and EDA showed a strict correlation with the thermal sensation of subjects. In Refs. [20,42] studies a linear relationship between ST and environmental temperature changes has been demonstrated. Also, for EDA and thermal comfort studies, the results revealed a positive degree of correlation between thermal comfort and EDA signal, and its dependency on the environmental conditions [43–45].

Within this framework, this study represents an experimentally driven original contribution towards the holistic and quantitative decoding of human comfort in the built environment through physiological measurements, which are basically driven by comfort-related

perception. The proposed procedure consists of the simultaneous collection of multiple physiologic signals, such as (i) EEG, (ii) HRV, (iii) ST, and (iv) EDA for investigating the thermal sensations. Recruited subjects were thus exposed to preliminary designed thermal boundaries (representative of neutral, warm, and cold conditions, clearly distinct) within an environmentally controlled and monitored laboratory facility. The experimental campaign took advantage of mostly commercial wearable systems already available in the market and characterized by low invasiveness bias on the tested subjects, for their physiological monitoring: the Muse2-Bundle, a commercial headband designed as a meditation-aiding device, collected 4-channels EEG [46]; the Empatica E4, a medical-grade wearable wristband device, measured HRV, ST and EDA signals [47]. Pre-processing and statistical analysis procedures were applied to identify the main physiological indicators of the human thermal perception, useful to improve the existing PCMs.

The structure of this paper is organized as follows. Section 2 reports about the physiological background. Section 3 refers to the novel proposed research methodology. Section 4 presents the study results, highlighting the main physiological differences of human thermal perception related to different environmental temperatures. Section 5 discusses the limits of the presented methodology and future development, and Section 6 concludes this study.

2. Theoretical background

2.1. Physiology of the brain–heart interactions

The interaction between brain and heart activities has been largely studied, especially for a better understanding of mutual physiological mechanisms. The scientific field dedicated to brain-heart communication, named neuroradiography [48], outlines the physiological networking of the nervous and cardiovascular systems [49]. Brain-heart communication is dynamic and two-way interactive; it occurs in four major ways: neurologically (through the transmission of nervous impulses), biochemically (via hormones and neurotransmitters), biophysically (through pressure waves), and energetically (through electromagnetic field interactions). The efferent (descending) pathway from the brain in the ANS, thought SNS and PNS, is integrated into the heart's intrinsic nervous system parallel to sensory neurons activities that detect pressure, HR, and hormones in the heart. The neural signal, from the intrinsic cardiac nervous system, propagates to the brain via afferent (ascending) pathways in both the spinal column and vagus nerves, reaching the cerebral cortex. In addition, the intrinsic cardiac nervous system has both short-term and long-term memory functions and can operate independently of central neuronal command. The biochemical communication occurs via the heart's hormones. The heart contains cells that synthesize and release catecholamines (norepinephrine, epinephrine, and dopamine), which are neurotransmitters once thought to be produced only by neurons in the brain and ganglia. More recently, it was discovered the heart also secretes the same range of oxytocin produced in the brain, commonly referred to as the love or social bonding hormone [50].

2.2. Physiology of the brain - skin interactions

Several studies have demonstrated how strictly brain activities and skin functions are connected [51]. The skin is associated to the peripheral sensory nervous system, i.e. ANS, and the central nervous system (CNS). The CNS communicates with the skin directly via efferent nerves and indirectly via adrenal glands [52]. Sensory as well as autonomic (sympathetic) nerves influence a variety of physiological functions (embryogenesis, vasocontraction, vasodilatation, body temperature, barrier function, secretion, growth, differentiation, cell nutrition, nerve growth). Fig. 1A illustrates brain-heart-skin nervous communication.

2.3. Thermoregulatory system

The human body is homeothermic, which means that the core temperature, in healthy and comfortable conditions, is kept nearly constant by physiological mechanisms [53]. The pivotal role in the thermoregulatory balance is represented by the hypothalamus, defined as the "body's thermostat". Changes in body internal temperature are recognized by thermoreceptors (central and peripheral) on the skin, which in turn transmit thermal afferent information to the CNS, where the thermal integration occurs. At this point, CNS sends the appropriate response via ANS (efferent pathway) to the target organs (e.g., skeletal muscle, skin vasculature, sweat glands), by triggering the thermal adaptation mechanisms. Heat conserving/generating or heat-dissipating

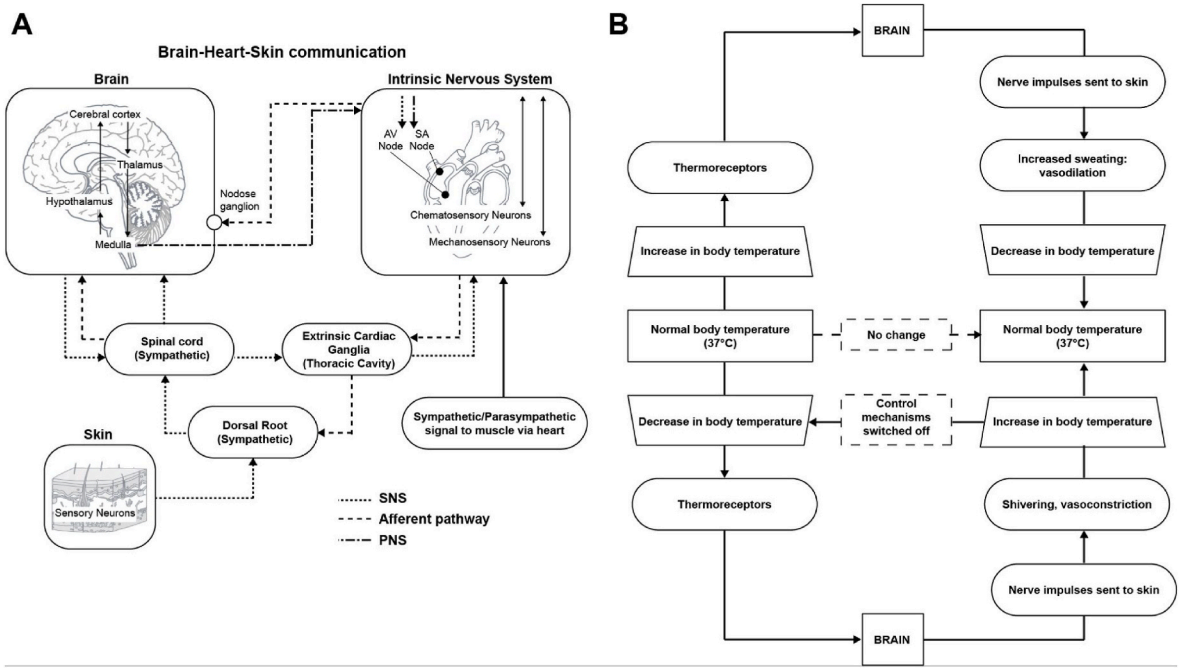


Fig. 1. (A) Brain-heart-skin communication pathways. (B) Physiological circuit of the thermoregulatory system.

reflexes to restrict or enhance heat loss. Heat-conserving is the main thermal mechanism to compensate for the reductions in the skin and/or body core temperature via peripheral vasoconstriction. During this process, blood vessels at the skin surface become narrowed to reduce heat dissipation throughout the skin. Shivering is the physical reaction during heat-conserving process, which consist of involuntary contraction of muscles. Cells' respiration releases heat to give energy for muscle contraction, which is used to warm up the body. When exposed to hot temperatures, the body temperature increases. In this case, the triggered thermal adaptation mechanism is the heat loss by vasodilation of blood vessels. Heat loss occurs by convection and radiation. The quantity of heat dissipated by the body is proportional to the amount of blood flow propagated through the skin. Vasodilation speeds up the heat loss process, the capillaries close to the most external layer of the skin dilate, and therefore more blood propagates, and the related transported heat power is dissipated. However, to enhance the amount of blood circulating via the superficial skin, the HR increases. Hence, the heart beats faster, which helps to increase the rate of heat loss from the body. Simultaneously, vasodilation reduces blood pressure. Then to restore normal blood pressure, the heart vehicles more blood through the blood vessels. Fig. 1B illustrates the physiological circuit of the thermoregulatory system.

3. Materials and methods

The designed experimental protocol aimed at clearly identifying the contribution of thermal comfort and sensation on physiological responses by exposing recruited subjects, individually taking part to the experiment, to three thermal environments (namely, neutral, cold, and warm) while being monitored through dedicated wearables in terms of their EEG, HRV, EDA, and ST signals. The three-test series differ in terms of internal operative temperature that was designed to be constant during each test and equal to the operative temperature recognized for thermally neutral environment, according to standards, and for significantly divergent thermal scenarios (almost 10 °C lower and higher with respect to the standard neutral one). The same procedure was replicated in both summer and winter 2021 and both multidomain effects and possible artifacts generally affecting EEG and HRV signals records were carefully controlled and limited throughout tests. Materials, experimental procedure, data processing, and statistical analysis implemented for achieving such goal are specified in the following subsections.

3.1. Experimental environment and equipment

The experimental sessions took place in the environmentally conditioned volume (4 × 4x2.7 m) of the NEXT.ROOM (Fig. 2A), a human multidomain comfort test room built at the Environmental Applied Physics Lab (eaplab.net) at the University of Perugia (Italy) [54,55]. The test room is a laboratory facility specifically designed for addressing gaps in human-comfort theories where environmental stimuli belonging to different comfort spheres are completely controlled and monitored [54]. The environmental monitoring setup is summarized in Table 1.

This highly controlled environment guarantees to limit the environmental variations among the three series of tests only to the ambient temperature while keeping constant other environmental stimuli like the illuminance level and the artificial lighting colour temperature, the CO_2 concentration, and the background noise level (only due to the conditioning systems operation). The internal layout consists of a table positioned in the middle of the room and a wheeling chair representing an office-like environment. The environmental conditions in the facility are controlled by means of combined conditioning systems (a radiative- and an air-based system), two artificial lighting systems - four light emitted diode (LED) panels, and two red-green-blue (RGB) reflectors -

Table 1
Environmental monitoring equipment installed in the NEXT.ROOM.

Measured parameter	Technical specifications	Position
Air temperature	Accuracy: ± 0.1 °C	Height: 0.10/0.60/1.10/1.60 m
Relative humidity	Accuracy: ± 1.5% (in 0 ÷ 90%)	Height: 1.10 m
Globe temperature	Accuracy: ± 0.1 °C	Height: 1.10 m
Superficial temperature	Accuracy: ± 0.1 °C	Four probes for each internal surface
Net radiation	Sensitivity: 10 μV/(W/m^2) Spectral range: 0.2 ÷ 100 μm	Orthogonal to the three dimensions of the test room
Air velocity	Accuracy: ± (0.2 m/s +3%)	Height: 1.10 m
Heat flux	Sensitivity: 60·10^{-6} V/(W/m^2)	One for each internal surface
Illuminance	Range: 20 ÷ 2000 lx	On the desk surface
CO_2 concentration	Accuracy: ± (50 ppm + 2%)	Height: 1.10 m

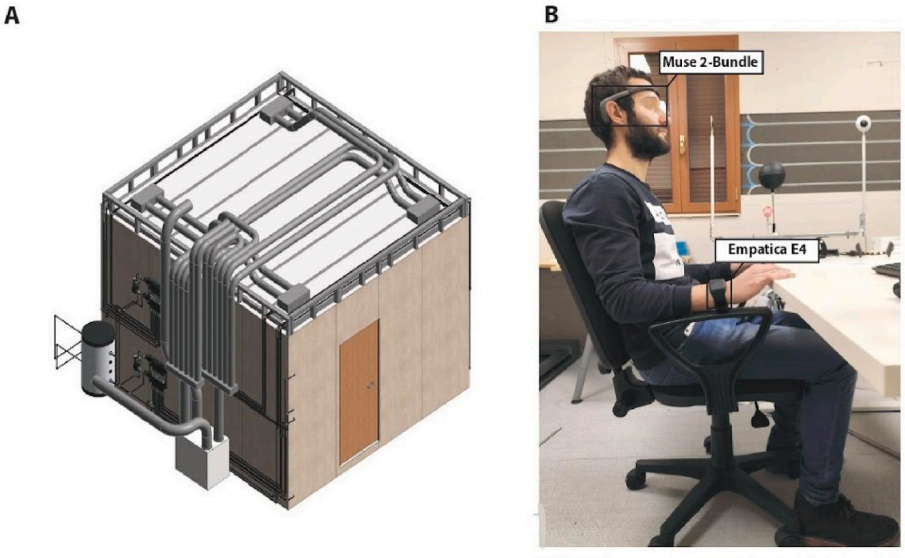


Fig. 2. (A) Building Information Modelling (BIM) representation of the climate chamber "NEXT.ROOM". (B) Measurements of physiological signals with Muse 2-Bundle and Empatica E4 during the experimental campaign.

and a mechanical ventilation system. The internal parameters, which are continuously monitored with a 5-min frequency rate, are the following: air temperature, relative humidity, globe temperature, superficial temperature, air velocity, CO_2 concentration, and illuminance level. A previous work by the authors presents more specifics about the facility microclimate control and environmental monitoring systems [55]. Despite its internal position, the NEXT.ROOM has a window aligned to the laboratory openings to provide an external view from inside the test room and to let natural daylight reach out the internal space. This window was closed during all the tests performed in this study in order not to expose subjects to different conditions in terms of natural daylight, local radiant asymmetry, or to provide them with diversified and uncontrolled visual stimuli. The four LED panels on the ceiling provided a uniform illuminance level on the desk surface of about 570 lx. The desired ambient temperature values were reached and kept constant throughout the whole duration of each test, thanks to the combined contribution of the installed Heating Ventilation and Air Conditioning (HVAC) and radiant systems. Moreover, the test room was always conditioned at least 2 h prior the experiments begin to verify the assumption of air temperature equal to the mean radiant temperature, and thus to the operative temperature that is the referring parameter for the identification of thermally neutral conditions in ISO 7730 [56]. Once in the NEXT.ROOM, each participant was asked to wear the two devices selected for the physiological signals recording: the Muse 2-Bundle (InterAxon Inc.), and the E4-wristband (Empatica) (Fig. 2B).

3.2. Physiological measurements

A commercial, low invasive, EEG sensor (Muse 2-Bundle, InterAxon Inc.) [46] was used for the electroencephalographic (EEG) recording. Several studies validated the usage of this low-cost portable device for continuous recording of EEG data [57,58]. In particular, a preliminary study was conducted [28] to validate the presented experimental protocol and the MUSE's capability of performing thermal comfort assessment. The study demonstrated the accuracy of the device in collecting the EEG and its low invasiveness. Among the measuring points of the International 10/20 system of the electrode placement [24], the device acquires brain activity from 4 measurement points. The reference electrode FPz (CMS/DRL) is located on the forehead, the two input electrodes are in front (left and right of the reference: AF7, AF8, silver made) and the other two are posterior, above each ear (TP9 and TP10, conductive silicone -rubber, Fig. 3). The device acquires signals at 256 Hz sampling frequency. The applicability of the EEG sensor is quick and simple, due to the presence of dry electrodes. EEG data were collected using the MUSE application [59], paired with a smartphone through Bluetooth Low Energy (BLE).

The Empatica E4 [47] (Fig. 4) is a multi-sensor bracelet for continuous monitoring of physiological signals. It has an internal flash memory

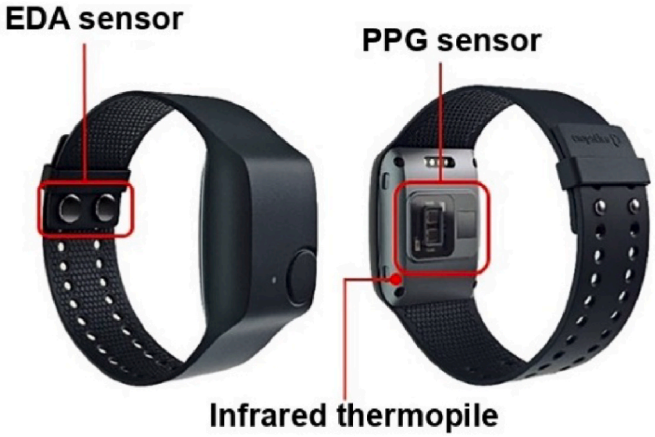


Fig. 4. Empatica E4 wristband, an overview of sensors.

to store data collected and streaming capacity through a BLE. The sensors are placed at the bottom of the device, in firm contact with the skin. Empatica E4 can register the following physiological signals:

- *BVP:* PPG polycarbonate (LG-SC1004A) technology captures the variation of volume of arterial blood under the skin resulting from the heart cycle. Two LEDs generate light at different wavelengths (Green and Red) and two photodiodes calculate the reflected light. BVP sampling frequency is 64 Hz. Sensor output resolution 0.9 nW/Digit, sensitive area: 14 mm^2.
- *IBI:* The IBI sequence is obtained from the processing of the BVP signal, with a process that removes incorrect peaks due to noise in the BVP signal.
- *EDA:* Two electrodes snap-on silver (Ag) plated with metallic core measure the constantly fluctuating changes of skin electrical properties. The EDA signal is captured at sampling frequency of 4 Hz, with a resolution of 1 digit ~900 pSiemens.
- *ST:* An infrared Thermopile reads peripheral ST at 4 Hz sampling frequency. Its resolution is 0.02 °C and the accuracy ±0.2 °C.

3.3. Experimental procedure

Before starting the whole measurement campaign, the calculation of the effective sample size was performed using the G*Power3 statistical test [60]. In a priori power analysis of G*Power3, the sample size was computed setting the required power level (1- β, i.e., Type II error probability of falsely retaining an incorrect null hypothesis H_0) equal to 0.95, the prespecified significant level (α, i.e., Type I error probability) equal to 0.05, the effect size f (which is the degree of deviation from H_0

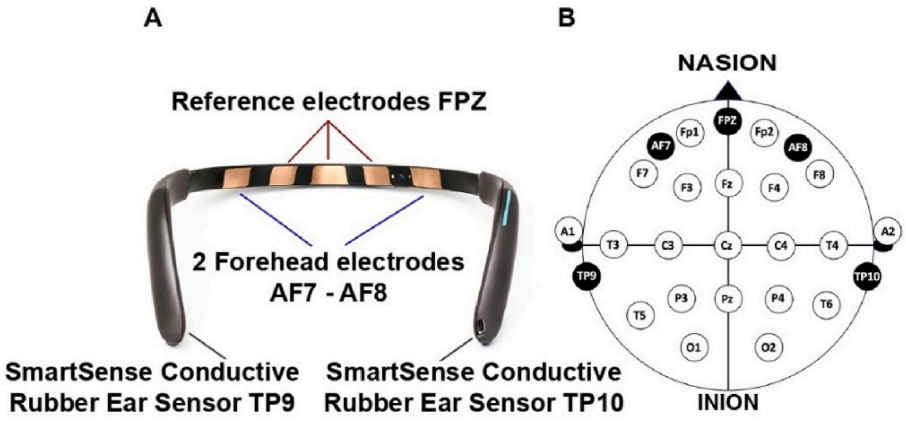


Fig. 3. (A) MUSE 2 headband sensors overview. (B) Top-down view of the EEG electrode positions on the subject's head.

in the underlying population H_1) equal to 0.25, and statistical test on repeated measures - within factor ANOVA and test Wilcoxon signed-rank test. The obtained sample size was 43 and 47 for the parametric and non-parametric tests, respectively.

The recruited sample group was made by 52 volunteers, 29 females (Table 2). Twenty-one volunteers involved in the experimental campaign repeated the same test by three times per season (for a total of 126 significant tests), and thirty-one subjects repeated the test by three times in a single season (for a total of 93 significant tests). The three tests were designed with three different and pre-defined operative temperatures, as follows: (i) 25 °C and 23 °C, in summer and winter, respectively, as representative of the thermally neutral scenario in accordance with ISO 7730:2005 [56] by assuming a clothing resistance level typical for each season (i.e., 0.5 and 1 for summer and winter seasons, respectively), and a low metabolic rate (i.e., equal to 58 W/m^2 for sedentary activities); (ii) 16 °C and 18 °C during the winter and summer tests representative of the cold scenario; (iii) 33 °C and 35 °C during the winter and summer tests representative of the warm scenario. These operative temperature extremes guaranteed that most of the participants would feel cold or warm during the tests so that variations in measured physiological signals with respect to the neutral benchmark could be reasonably associated to changes in thermal sensation and not to other confounding factors.

Each subject took part in the experiment individually. Tests were scheduled from 10:00 a.m. to 1:00 p.m., and from 3:00 p.m. to 6:30 p.m. Subjects were instructed not to smoke, perform any physical intense effort, not to eat or drink coffee since at least 1 h before their scheduled tests. Each test duration corresponds to 20 min (Fig. 5).

Before entering the test room, one of the authors explained to the participant the broad scope of the experiment, i.e., to enhance our knowledge on human comfort, so that each subject provided a written informed consent and thus filled up a general survey on personal information including age, gender, height, and weight. Once positioned in the NEXT.ROOM, each subject has been spending 15 min for personal acclimatization before starting the 5-min physiological signals recording. During the 5-min physiological recording, the subject was instructed to stay as quiet as possible, without performing any specific task, just sitting in the middle of the room, and relaxing to keep their metabolic rate constant and reduce measurement artifacts due to body movements. At the end of the test, the subjects filled out a second survey providing their thermal environment evaluation on a 5-points Likert scale ranging from +2, meaning warm sensation, down to −2, meaning cold sensation, where the 0 represents neutrality. The scale was designed in accordance with the purpose of the study that aimed at stressing individuals on extreme sensations achievable indoors to clearly distinguish the three main conditions of thermal neutrality, or warm/cold sensation. Despite each thermal scenario was experimentally designed to be clearly perceivable as neutral, cold, or warm, the filling out of the survey allowed to account for subjects' diversity in perceiving the thermal environment.

3.4. Physiological signals processing

Collected physiological signals were processed to extract a total of 110 features (85 in the frequency-domain, and 25 in the time-domain; summarized in Table 3 and Table 4).

Table 2
Recruited sample group description.

Variable	Average ± standard deviation	Minimum	Maximum
Age [years old]	28.6 ± 3.9	20	39
Body Mass Index [kg/m^2]	22.1 ± 3.6	16.8	31.3

3.4.1. EEG pre-processing and features extraction

A variety of tailored procedures were put in place for data collection, signal pre-processing, and feature extraction of the three main collected physiological signals. Concerning the EEG, an on-board Digital Signal Processing (DSP) module of Muse allowed noise filtering (0.1–45 bandpass and notch at 60Hz), eye blink, and jaw clench artifacts removal. Fast Fourier Transform (FFT) was applied to the raw data using a 256-samples window, with a step of 22 samples. The PSD of five frequency bands was obtained, each of those corresponding to a particular mental state. Delta waves (0.5–4 Hz) are related to the deep sleep state, theta waves (4–8 Hz) are associated with deeply relaxed and inward-focused states, alpha waves (8–12 Hz) are dominant in very relaxed and passive attention, beta waves (12–35 Hz) correspond to an active mind and external attention states, and gamma waves are prominent in concentration and problem solving conditions [61]. The average power in these bands is real-time computed in hardware and sent alongside an indicator of data quality from each sensor. In addition, for each of the five-frequency bands, across the 4 considered channels (AF7, AF8, TP9, TP10) the relative power was calculated using equation (1), where B represents the power of the signal in a specific frequency band and T represents the total power:

$$Relative\ EEG\ band\ power = \sum_{0}^{i} B \Big/ \sum_{0}^{i} T \tag{1}$$

Besides the relative power calculation, the band ratio (delta/beta; delta/gamma; theta/beta; theta/gamma; alpha/beta; alpha/gamma), and alpha per beta in each channel was also calculated. Measures of frontal (2) and temporal (3) asymmetry of each band were computed separately by subtracting band power in the left hemisphere from power at homologous sites in the right hemisphere [25]:

$$Frontal\ asymmetry = B\mathrm{AF8} - B\mathrm{AF7} \tag{2}$$

$$Temporal\ Asymmetry = B\ \mathrm{TP10} - B\mathrm{TP9} \tag{3}$$

The total number of calculated EEG features was 78.

3.4.2. HR and HRV pre-processing and features extraction

Improper wear of the HRV-sensing bracelet and movement of subjects are the main sources of artifacts in heart activity signals recorded by the device. The HRV analysis [62] (python module for HRV analysis) was used for processing the heart activity signal. This package was useful to remove outliers and ectopic beats from the inter beat interval (IBI) signal. Sixteen HR-HRV's time-domain and seven frequency-domain features from free-artifacts IBI were calculated. Time-domain variables express the variability in HR or the intervals between recurring normal cardiac cycles. PSD analysis instead, provides evidence of how power distributes as a function of frequency [63]. The most representative features extracted were selected from literature, according to Refs. [64,65] Table 3.

3.4.3. Skin parameters pre-processing and features extraction

As regards EDA, it is worth mentioning that the skin conductance (SC) signal is hypersensitive to physical activity and temperature variations. In this view, it is essential to filter them out from signals. EDA toolkit 11 [66] for artifacts detection was used. Portions of data identified by this toolkit as artifacts were manually removed from the signals. Eight EDA features (Table 4) were obtained from artifact-free signals. They are extracted from both the phasic and tonic components of the EDA signal. The phasic signal of EDA or skin conductance response (SCR) represents the change in the wetness level of the skin in response to a presentation of a stimulus. On the contrary, the tonic component or skin conductance level (SCL) is the average of conductivity given by sweat production during a time interval. The cvxEDA tool 13 [67] was used to decompose the EDA signal into its two components. This tool uses convex optimization to predict the activity of the ANS by

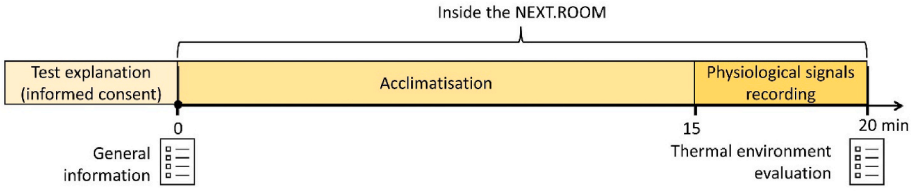


Fig. 5. Single test procedure.

Table 3
HR time domain, HRV time and frequency-domain measures. EDA features of tonic and phasic components.

	Features	Units	Description
HRV time-domain	**NNI Mean**	ms	The mean of normal-to-normal beat intervals (R-R intervals).
	SDNN index	ms	The standard deviation of the time interval between successive normal heartbeats
	SDSD	ms	Standard deviation of differences between adjacent NN intervals
	RMSSD	ms	The square root of the mean of the sum of the squares of differences between adjacent NN intervals
	NNI Median	ms	Median Absolute values of the successive differences between the RR-intervals
	NNI 50	ms	Number of interval differences of successive RR-intervals greater than 50 ms
	NNI 20	ms	Number of interval differences of successive RR-intervals greater than 20 ms
	Range NNI		Difference between the maximum and minimum NNI interval
	CVNNI	ms	Coefficient of variation equal to the ratio of SDNN divided by NNI Mean
	pNN50	%	The proportion derived by dividing NNI 50 (The number of interval differences of successive RR-intervals greater than 50 ms) by the total number of RR-intervals
	pNN20	%	The proportion derived by dividing NNI 20 (The number of intervals differences of successive RR-intervals greater than 20 ms) by the total number of RR-intervals
	CVSD	ms	Coefficient of variation of successive differences equal to the RMSSD divided by NNI Mean
HR time-domain	**HR Max**	bpm	Max HR
	HR Min	bpm	Min HR
	HR Mean	bpm	Mean HR
	STD HR		Standard deviation of HR
HRV frequency-domain			***Analysis of short-term recordings (5 min)***
	VLF		Power in VLF range (≤0.04 Hz)
	LF		Power in LF range (0,04–0,15 Hz)
	LF_{norm}	nu	LF power in normalized units LF/(total power-VLF) × 100
	HF	ms^2	Power in HF range (0.15–0.4 Hz)
	HF_{norm}	nu	HF power in normalized units HF/(total power-VLF) × 100
	LF/HF		Ratio LF (ms2)/HF (ms2)
	Total power	ms^2	Total power density spectral

Table 4
Skin parameters: ST and EDA tonic and phasic features.

	Features	Units	Description
Tonic component	**Mean Tonic**	mS	Mean value of tonic component
	Tonic perc25	mS	20th percentile
	Tonic perc75	mS	75th percentile
	Tonic perc80	mS	80th percentile
	Tonic SD	mS	Standard deviation
	Tonic quartdev	mS	Quartile deviation
Phasic component	**Phasic Perc20**	mS	20th percentile
	Max Peak	mS	Max Peak of phasic component
T_{sk}	**Mean ST**	$^{\circ}C$	Mean skin temperature

means of Bayesian statistics. The mean value of skin temperature was also calculated.

3.5. Statistical analysis

To establish the correlation between physiological features and the thermal sensations of participants, a dedicated statistical analysis was performed. The physiological features, calculated from measured data, were classified according to the thermal sensation (TS) expressed by the subjects in the questionnaires. Categories corresponded to C: cold (TS $<$ 0), N: neutral (TS $=$ 0), and W: warm (TS $>$ 0) groups. Data points with Z-score higher than 2 were considered outliers and removed from the dataset. Mean ($\bar{f}$) and standard uncertainty of the mean (u_f) of each feature was then calculated according to the Guide to the expression of uncertainty in measurement (GUM) [68]. The normality and the homogeneity of variance, for each group of features, were evaluated with the Shapiro test [69] and Bartlett test [70], respectively. The repeated measures ANOVA model [71] was used to compare means of physiological features across variables based on repeated observations in different conditions, when the features satisfied the assumptions of normality and homogeneity of variance (in this case: Alpha beta ratio TP10, Beta TP10, Beta TP9, Relative delta TP10, Relative gamma TP9, CVSD, HF_{norm}, LF_{norm}, HR Mean, RMSSD, SDSD), in opposite Kruskal-Wallis non-parametric test [72] was performed (Gamma TP10, Relative alphaTP9, Relative gamma TP10, HF, LF/HF, HR Mean, NNI Mean, NNI Median, HR min, pNN20, pNN50, SDNN, Tonic mean, Tonic perc25 and the skin feature Mean ST). The capability of the features to discriminate between C–N, C–W, and N–W thermal sensations was further investigated with a post-hoc analysis based on the Dwass-Steel-Critchlow-Fligner pairwise comparison test [73]. Therefore, we pointed out statistically significant variations of each feature depending on personal thermal perception. The above-described statistical analysis was performed before and after the cross-validation test. This specific test was designed to solve ambiguities in subjects' thermal evaluation since 6 subjects exposed to the cold temperature and 19 subjects exposed to the warm temperature expressed TS $=$ 0 (neutrality). For simplicity, $X_{(C-N)}$ was used to label the group of subjects who expressed a feeling of thermal neutrality during the cold test and $X_{(N-W)}$ for those who expressed a feeling of thermal neutrality during the warm test. Gamma TP10, normalized low frequency (LF_{norm}), low to high frequency ratio (LF/HF), and normalized (HF_{norm}) were used for the validation, being the most statistically significant features, assessed by the first computation of the Dwass-Steel-Critchlow-Fligner test - Gamma TP10 in C–N contrasts, and LF_{norm}, LF/HF, and HF_{norm} in N–W contrasts. Considering the non-gaussian distribution of the selected features, the Mann-Whitney U non-parametric *t*-test between [$X_{(C-N)}$- N], [$X_{(C-N)}$- C], [$X_{(N-W)}$- W] and [$X_{(N-W)}$ - N] groups were implemented. The results revealed that there is a statistical difference between the group $X_{(C-N)}$

and C, and between $X_{(N-W)}$ and W groups. The corresponding features of those subjects were then categorized as N. Table 5 summarizes the cross-validation results.

4. Results

Regarding the indoor environmental condition during the experimental sessions, the standard deviation of the observed air temperature data was always below 0.4 °C during each test. Additionally, the internal CO_2 concentration (also continuously monitored) was always kept below 500 ppm thanks to the mechanical ventilation operation. Statistical results of physiological responses are summarized in Table 6 and Table 7, and shown in Fig. 6A and Fig. 7.

Regarding the EEG physiological domain, the results partially confirmed what was found in the previous study [28]. In general, EEG measurements showed a correlation with the thermal sensation in terms of increase or decrease of power of brain waves. The warm sensation could be correlated to an increase of relative alpha. On the other hand, cold sensation is correlated to an increase of beta and gamma power. Beta and Gamma TP10, Relative alpha TP9 and Relative gamma TP10 showed a statistical difference between cold and warm sensations; Gamma TP10 also showed a difference between cold and neutral thermal sensations. Considering the HRV domain results, the time-domain HRV features related to the normal-to-normal beat intervals (NNI) were higher in warm than in cold scenarios. Among the HRV frequency-domain features, HF and HF_{norm}values were larger under lower temperatures. On the contrary, normalized LF_{norm} showed an increase of power under higher temperatures. LF/HF was higher in warm than cold conditions. All the HRV investigated features present significant differences between cold and warm thermal sensation. HF_{norm}, LF_{norm}, and LF/HF showed also differences between neutral and warm sensations. The tonic component of the EDA signal and the mean ST were identified as the most statistically significant skin features. High values of mean ST and EDA tonic perc25 were found in the warm state. The skin features showed significant differences between cold and warm sensation, the EDA Tonic perc25 also between cold and neutral, the mean ST also between neutral and warm. Among the four measured physiological signals, Gamma TP10, LF/HF and the tonic component of EDA present the strongest significant and consistent differences between cold and warm thermal sensation. Their combination can be therefore used to uniquely identify cold or warm personal thermal states (Fig. 6B) as never done before.

Now, the observed findings are interpreted according to human physiology described in Section 2. As the brain, heart, and skin are physiologically connected by ANS communication, the combination of EEG, HRV, EDA, and ST signals analysis is a reasonable approach to uniquely identify the thermal sensations of the subjects. The dominance of an EEG frequency band is correlated to a certain mental state. In general, low-frequency bands such as delta, theta, and alpha correspond to deep sleep, deeply relaxed, and very relaxed conditions, respectively. On the other hand, high frequency bands (beta and gamma) are prominent in active mind, concentration, and problem-solving tasks. Previously conducted studies, aimed at highlighting the potential correlation between the prevalence of a specific EEG frequency band and the perceived thermal sensations, revealed that EEG alpha waves increased while beta waves decreased with a pleasant thermal state, others showed that high values of alpha power correspond to a high state of arousal level and vice versa. The outcomes of this study showed that relative alpha increased when subjects felt warm thermal sensation. Therefore, when subjects are exposed to a warm environment, they were observed to be better relaxed, less focused, and unable to keep their attention. On the contrary, in cold conditions, there is an increase in the activity of beta and gamma bands (>35 Hz), which are the main brain waves connected to intense mental activity, as clearly depicted in Fig. 8A (referring to a single subject selected as representative of the tested sample). As confirmation of our method reliability, these outcomes are in line with previous results in literature as summarized in the review by the same authors [27].

In addition, it was found that, while the HR of the human body increases under warm temperature due to the thermoregulatory mechanism working for heat dissipation, consequently, the NNI intervals decrease. Among the frequency domain features derived by PSD analysis (Fig. 7B), HF was found to be higher when subjects felt cold than warm sensation. In the meantime, higher values of LF and LF/HF were found when subjects were exposed to a higher temperature. In general, HF, or the respiratory band, is affected by breathing from 9 to 24 bpm and reflects the PNS activity, which is involved in the relaxing phase, it allows the body to rest and recover, decreasing the HR. LF band, instead, is produced by PNS and SNS and blood pressure (BP) regulation via baroreceptors. In the resting state, the LF band manifests baroreflex activity and not cardiac SNS drive. The LF/HF power was originally defined as the expression of the "PNS-SNS balance". However, many studies [74] suggest that LF/HF cannot be the indicator of the PNS-SNS balance, considering that PNS and SNS interaction is complex, nonlinear, and non-reciprocal, and the LF power is not a pure index of SNS activity.

In this view, showing higher values of LF and lower values of HF in warm conditions could be explained by different characteristics in ANS and PNS excitation. The sympathetic branch of the ANS got excited under cold environments to constrict the blood vessels in the skin, and therefore reducing personal heat dissipation. Under hot environments, however, it is excited to make the body sweat, dissipating the extra heat. Normally, the intensity of the excitation of the sympathetic nerve is much higher when the human body sweats. Consequently, the LF/HF had higher values when the human body was exposed to warm environments.

Moreover, also the mean ST tends to increase as the environmental temperature increases. Nevertheless, a meaningful difference between cold and neutral states is not clearly detectable (Fig. 8). Physiological response of the skin, measured by EDA, revealed information about the electrical conductance of the skin, itself, which varies with emotion and thermal changes. Sweat glands are controlled by the sympathetic nervous system, in this view the skin conductance can be informative about the psychological or physiological arousal level of the human body. The moisture level depends on the sweat glands and blood flow, which are regulated by the SNS. The results showed that subjects under warm conditions present a higher value of the tonic component of EDA because of high activation of the SNS (Fig. 8D), responsible for the activation of sweat glands during heat dissipation mechanism.

Nevertheless, results do not provide clear information about which physiological features combination clearly distinguish the neutral thermal sensation. This outcome could be justified by differences in individual subjects' thermal preferences. In fact, it is recognized that under the same environmental conditions different people can perceive different thermal sensations [17]. In the presented study we classified data according to the thermal sensation expressed by subjects, and by using cross-validation method we reorganized the data redirecting the

Table 5
The Mann-Whitney U non-parametric *t*-test results (*for p-value < 0.05). A non-parametric test was implemented since the used features showed a non-gaussian distribution.

Features	Groups	t-value	p-value
Gamma TP10	$X_{(C-N)}$- N	101	0.06
	$X_{(C-N)}$- **C**	**86**	**0.01***
LF_{norm}	$X_{(N-W)}$- N	577	0.5
	$X_{(N-W)}$- **W**	**410**	**0.004***
LF/HF	$X_{(N-W)}$- N	550	0.4
	$X_{(N-W)}$- **W**	**452**	**0.01***
HF_{norm}	$X_{(N-W)}$- N	578	0.6
	$X_{(N-W)}$- **W**	**412**	**0.003***

Table 6
Repeated measured ANOVA - Kruskal-Walli's test results. The mean feature ($\bar{f}$), the standard uncertainty of the mean (u_f), H-statistic, and the significant p-value (p-value < 0.01) are reported for each statistically significant feature extracted by the retrieved physiological signals.

	Features	$\bar{f} \pm u_f$ (C)	$\bar{f} \pm u_f$ (N)	$\bar{f} \pm u_f$ (W)	H-statistics	p-value
EEG	Alpha beta ratio TP10	1,2 ± 0,06	1,3 ± 0,06	1,3 ± 0,06	7,2	<0,001
	Beta TP10	0,6 ± 0,03	0,5 ± 0,03	0,5 ± 0,03	12,6	<0,001
	Beta TP9	0,5 ± 0,04	0,5 ± 0,03	0,5 ± 0,04	6,9	<0,001
	Gamma TP10	0,2 ± 0,04	0,1 ± 0,03	0,1 ± 0,04	17,6	<0,001
	Relative alpha TP9	0,3 ± 0,01	0,3 ± 0,01	0,3 ± 0,01	16,0	<0,001
	Relative delta TP10	0,2 ± 0,01	0,2 ± 0,01	0,2 ± 0,02	6,7	<0,001
	Relative gamma TP10	0,1 ± 0,02	0,1 ± 0,01	0,1 ± 0,01	11,8	<0,001
	Relative gamma TP9	0,1 ± 0,02	0,0 ± 0,02	0,0 ± 0,02	5,7	<0,001
HRV	CVSD	0,1 ± 0,03	0,1 ± 0,003	0,0 ± 0,03	26,2	<0,001
	HF	608,1 ± 73,14	507,1 ± 66,91	393,4 ± 57,34	18,6	<0,001
	HF_{norm}	39,3 ± 3,8	36,9 ± 3,27	27,9 ± 2,64	26,1	<0,001
	LF/HF	1,7 ± 0,23	2,1 ± 0,38	3,1 ± 0,43	26,0	<0,001
	LF_{norm}	60,7 ± 3,8	63,1 ± 3,27	72,1 ± 2,64	26,1	<0,001
	HR Max	85,4 ± 2,27	90,2 ± 2,54	91,1 ± 2,55	10,5	<0,001
	HR Mean	68,6 ± 2,26	71,7 ± 2,34	74,4 ± 2,18	11,0	<0,001
	NNI Mean	896,6 ± 30,18	843,3 ± 29,41	813,4 ± 24,74	13,9	<0,001
	NNI Median	900,3 ± 30,86	843,8 ± 29,91	812,9 ± 24,97	15,4	<0,001
	HR Min	56,1 ± 1,85	59,9 ± 1,94	61,1 ± 2,06	9,8	<0,001
	pNN20	64,4 ± 2,72	58,7 ± 3,31	56,1 ± 2,68	15,4	<0,001
	pNN50	27,9 ± 3,54	20,7 ± 2,81	16,1 ± 2,69	24,1	<0,001
	RMSSD	53,4 ± 3,93	45,5 ± 3,10	40,7 ± 3,04	25,1	<0,001
	SDNN	62,4 ± 4,19	55,9 ± 3,82	53,3 ± 3,12	10,7	0,005
	SDSD	53,4 ± 3,92	45,5 ± 3,10	40,7 ± 3,04	25,1	<0,001
EDA	Tonic mean	−1,0 ± 0,22	−0,6 ± 0,12	−0,5 ± 0,08	9,5	0,009
	Tonic perc25	−1,6 ± 0,22	−1,2 ± 0,12	−1,0 ± 0,08	24,5	<0,001
T_{sk}	Mean ST	31,3 ± 0,33	32,1 ± 0,48	33,2 ± 0,51	27,2	<0,001

Table 7
Dwass-Steel-Critchlow-Fligner pairwise comparison results. Statistical p-values obtained by contrast between C–N, CW, N–W (* for p-value < 0.01).

	Features	C–N (p-value)	C–W (p-value)	N–W (p-value)
EEG	Alpha beta ratio TP10	0,2	0,01	0,8
	Beta TP10	0,02	**0,003***	0,7
	Beta TP9	0,02	0,02	0,9
	Gamma TP10	**0,003***	**0,001***	0,7
	Relative alpha TP9	0,01	**0,001***	0,5
	Relative delta TP10	0,8	0,1	0,4
	Relative gamma TP10	0,03	**0,003***	0,6
	Relative gamma TP9	0,1	0,02	0,8
HRV	CVSD	0,02	**0,001***	0,04
	HF	0,1	**0,001***	0,04
	HF norm	0,6	**0,001***	**0,001***
	LF/HF	0,7	**0,001***	**0,001***
	LF norm	0,6	**0,001***	**0,001***
	HR Mean	0,2	**0,002***	0,3
	NNI Mean	0,04	**0,001***	0,5
	NNI Median	0,03	**0,001***	0,4
	pNN20	0,1	**0,001***	0,3
	pNN50	0,01	**0,001***	0,1
	RMSSD	0,01	**0,001***	0,1
	SDNN	0,04	**0,004***	0,8
	SDSD	0,01	**0,001***	0,1
EDA	Tonic mean	0,03	0,01	0,9
	Tonic perc25	**0,006***	**0,001***	0,1
T_{sk}	Mean ST	0,1	**0,001***	**0,003***

subjects with ambiguities (e.g., expressed warm thermal sensation under neutral thermal conditions to the group where their physiological features statistically belong. Such procedure resulted in more data classified as warm or cold with respect to those classified as neutral. Only a few physiological features better distinguish the neutrality from cold or warm thermal sensation (e.g., Gamma TP10, Tonic perc25 in C–N contrast, LF_{norm}, HF_{norm}, LF/HF and Mean ST in N–W), which could mean that they are not sufficient to clearly describe in terms of physiological data the neutral sensation.

5. Discussion

We developed and experimentally demonstrated the use of physiological measurements for thermal comfort decoding, in different seasonal and thermal microclimate conditions. Combining synchronous analysis of four physiological signals, EEG, HRV, ST, and EDA for the first time, we provided a novel approach in the PCM field. Using low invasive wearable devices for acquiring physiological signals, and analysing data with ad hoc processing procedure, we uniquely identified cold and warm individual thermal sensation based only on physiological measurements. We believe the presented methodological approach can massively improve the implementation of a more robust personalized thermal comfort modelling and measurement, pointing at the optimization of buildings' energy consumption for HVAC operation. The outcomes of this study will be strategic in the field of thermal comfort investigations in different types of indoor environments and for several scopes, as human-building interaction decoding. To this aim, known limitations should be considered and overcome. As a matter of fact, the implemented experimental procedure was specifically designed for reducing the impact of non-thermal stimuli and body's movement artifacts in physiological signals measurement. The experiments were carried out in an environmentally controlled multidomain test room and with behaviourally controlled participants, while real-life scenarios would imply subjects performing a variety of activities being exposed to a simultaneous combination of a variety of physical boundaries (i.e., thermal, visual, acoustic, air quality stimuli). Therefore, the data acquisition scenario adopted for this work could be assumed as representative but a particular case of the real-life conditions. Nevertheless, different data analysis procedures to increase signal-noise-ratio are available, but further data processing is required for deep investigation. Among the acquisition tools limitations there are the invasiveness of wearable devices, and their accuracy. This is the reason why wearable devices have been selected to be easy-to-wear. Despite that, the EEG headband may cause headaches, which means that it is not suitable for a long period of acquisition. In addition, the presence of dry electrodes,

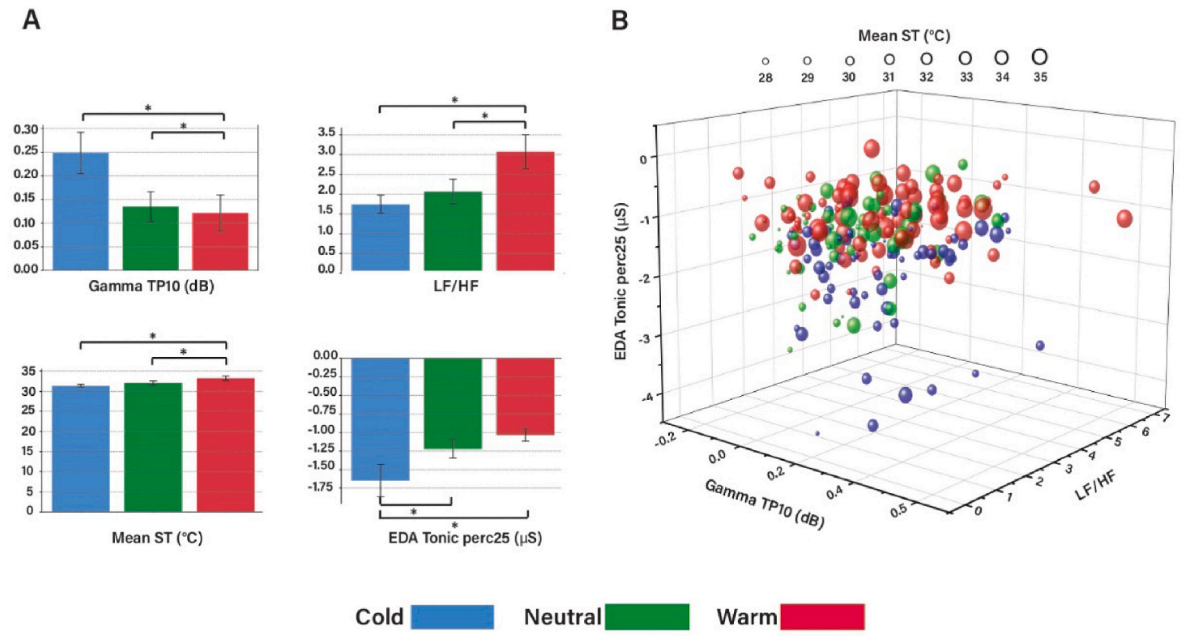


Fig. 6. (A) The mean ($\bar{f}$), and the standard uncertainty of the mean (u_f) of the main statistically significant features, (*P < 0.001). (B) Warm, neutral, and cold sensations as described by the combination of the most representative features extracted by EEG (Gamma TP10), HRV (LF/HF), EDA (Tonic perc25), and ST processing.

make the device more prone to wear out under particular conditions or after a certain time, risking compromising the data collection if the metrological characteristics are not periodically verified by the experiment manager. Finally, it should be mentioned that physiological signal characteristics are associated to personal factors. In fact, individual psychophysiological conditions and personal feelings can influence the thermal perception. Therefore, additional surveys to collect emotional states of participants could be useful to obtain an accurate analysis of the perceived thermal sensations in relation to a larger set of control variables. In conclusion, moving from this research outcomes, further investigations about the neutral state estimation and the quantitative correlation between physiological features and individual thermal sensation need to be conducted. A possible solution to address the high individual variability issue could be the employment of data normalization methods by using, for example, the physiological data recorded in resting and thermally neutral state as a reference. Additionally, to further refine our ability in decoding human thermal perception, it could be valuable to continuously monitor subjects' physiological response also under thermal gradients thus verifying human physiology responsiveness to environmental changes.

6. Conclusions

The presented research aimed at clearly identify individual thermal perception through physiological signal processing. A new laboratory-based experimental protocol was developed and conducted to expose recruited subjects (52 people in total) to three thermal boundaries as representative of neutral, cold, and warm conditions in both summer and winter. Physiological data collection was pursued through commercially available wearables providing multiple signals (i.e., EEG, ECG, EDA, and ST). Results only partially confirm previous research outcomes in the field of thermal perception, highlighting the importance of collecting all the physiological signals, here simultaneously analysed for the very first time. In fact, each signal provides features whose combination supports the detection of extreme thermal sensation, i.e., cold and warm. For the EEG measurements, the results showed that the warm sensation is correlated to an increase of relative alpha, and the cold sensation is correlated to an increase of beta and gamma powers. In addition, Beta and Gamma TP10, Relative alpha TP9 and Relative gamma TP10 has a statistical difference between cold and warm sensations. Considering the HRV results, the time-domain HRV features were higher when subject felt warm than cold sensation. Among the HRV frequency-domain features, HF and HF_{norm} values showed greater values when participants felt cold thermal sensation; on the contrary, LF_{norm} showed an increase of power for warm thermal sensation. LF/HF was higher in warm than cold conditions. In general, all the HRV features has significant differences between warm and cold states. Among skin measurements, the results demonstrated that high values of mean ST and EDA tonic component were related to the warm state. Both ST and EDA features has differences between cold and warm sensations, the EDA Tonic perc25 also between cold and neutral, and the mean ST also between neutral and warm thermal sensations.

Furthermore, data processing and cross-validation test suggest that human thermal sensation can be detected with higher precision relying upon physiological signals, aside from the environmental boundaries. Research outcomes support the idea for the development of personal control systems for the built environment mainly (if not solely) driven by the single-occupant physiology in a human-centric perspective. Monitored environmental parameters seem not to add any information on people indoor comfort that cannot be already detected via personal human-centered parameters. Individuals, even if belonging to the same climate region and presenting similar cultural background, present a variety of thermal comfort thresholds leading to different perceptions within the same environmental boundaries while their physiological response associated with cold and warm thermal perception is the same. This discovery therefore opens the doors toward fully personalized environmental comfort models really tailored to each one's expectations, attitudes, and other temporary or permanent individual characteristics. The decoding of human thermal perceptions discloses the massive potential of triggering human subjects toward their personal wellbeing through dedicated tailored low-energy and low-cost strategies based on building interfaced control algorithms.

CRediT authorship contribution statement

Silvia Angela Mansi: Writing – review & editing, Writing – original

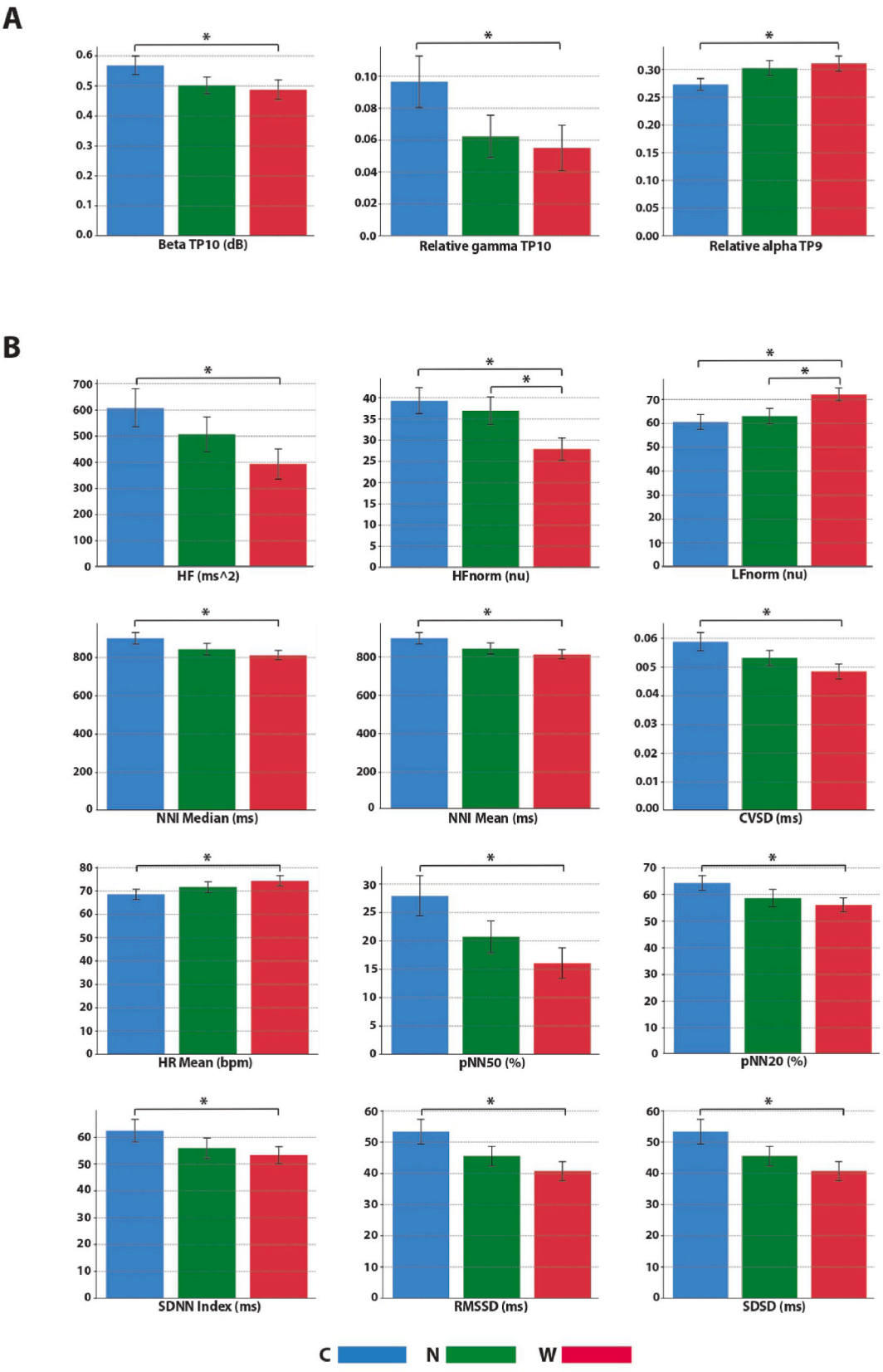


Fig. 7. Histograms of the Repeated measured ANOVA - Kruskal-Wallis test results. (A) The mean ($\bar{f}$) and standard uncertainty of the mean (u_f) of the statistical results concerning the EEG features (B) The mean ($\bar{f}$) and standard uncertainty of the mean (u_f) of the statistical results concerning the HRV features. *P < 0.001. (Bar colours: blue stands for C; green stands for N and red stands for W thermal sensations). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

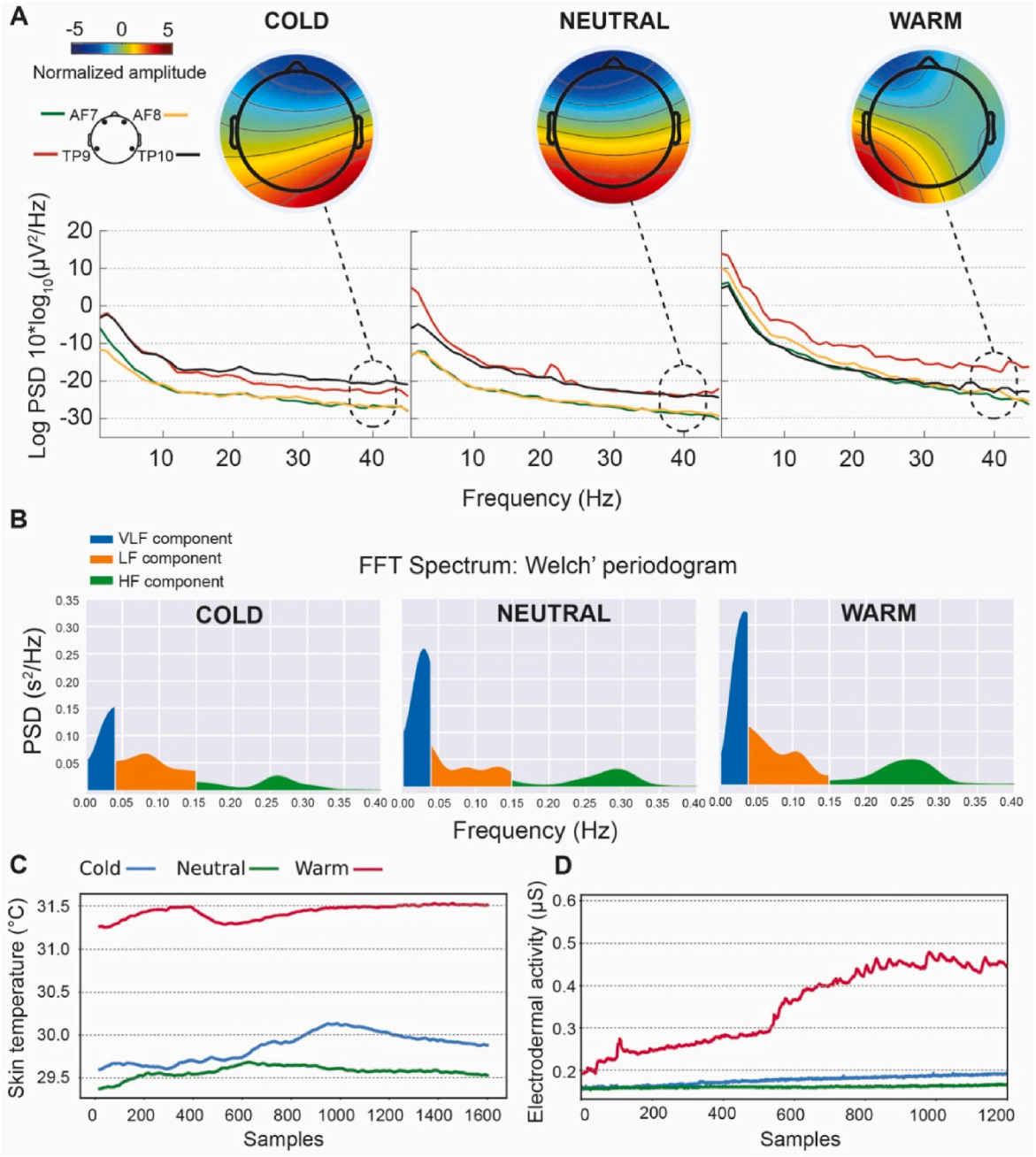


Fig. 8. Detailed results extracted from the analysis of the physiological signals measured on the same subject under three different thermal sensations: C, N and W. (A) PSD responses across all the EEG channels at 40 Hz. The scalp maps were generated using a ''topoplot'' function of EEGLAB (B) Welch' periodogram with PSD trends of HRV classified according to VLF, LF, and HF frequency bands. (C) Mean ST signal. (D) EDA signal.

draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ilaria Pigliautile:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Marco Arnesano:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Anna Laura Pisello:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgments

The authors wish to thank the Italian Ministry of research for supporting the young researcher PRIN project NEXT.COM (20172FSCH4_002). Additionally, Ilaria Pigliautile acknowledges the PON project (D.M. 1062 PON Ricerca e Innovazione) "Red-To-Green", and Anna Laura Pisello acknowledges the Horizon Europe ERC Grant HELIOS (GA 101041255).

References

[1] A. Heydarian, C. McIlvennie, L. Arpan, S. Yousefi, M. Syndicus, M. Schweiker, F. Jazizadeh, R. Rissetto, A.L. Pisello, C. Piselli, C. Berger, Z. Yan, A. Mahdavi, What drives our behaviors in buildings? A review on occupant interactions with building systems from the lens of behavioral theories, Build. Environ. (2020), https://doi.org/10.1016/j.buildenv.2020.106928.

[2] X. Xu, J.E. Taylor, A.L. Pisello, Network synergy effect: Establishing a synergy between building network and peer network energy conservation effects, Energy Build. (2014), https://doi.org/10.1016/j.enbuild.2013.09.017.

[3] Y.S. Tsay, R. Chen, C.C. Fan, Study on thermal comfort and energy conservation potential of office buildings in subtropical Taiwan, Build. Environ. 208 (2022), 108625, https://doi.org/10.1016/J.BUILDENV.2021.108625.

[4] Y. Al Horr, M. Arif, A. Kaushik, A. Mazroei, M. Katafygiotou, E. Elsarrag, Occupant productivity and office indoor environment quality: a review of the literature, Build. Environ. 105 (2016) 369–389, https://doi.org/10.1016/J.BUILDENV.2016.06.001.

[5] A.L. Pisello, C. Piselli, F. Cotana, Influence of human behavior on cool roof effect for summer cooling, Build. Environ. 88 (2015) 116–128, https://doi.org/10.1016/J.BUILDENV.2014.09.025.

[6] J. Wu, C. Liu, H. Wang, Analysis of Spatio-temporal patterns and related factors of thermal comfort in subtropical coastal cities based on local climate zones, Build. Environ. 207 (2022), https://doi.org/10.1016/J.BUILDENV.2021.108568.

[7] F. Stazi, F. Naspi, M. D'Orazio, A literature review on driving factors and contextual events influencing occupants' behaviours in buildings, Build. Environ. 118 (2017) 40–66, https://doi.org/10.1016/J.BUILDENV.2017.03.021.

[8] X. Zhang, X. Chao, L. Lou, J. Fan, Q. Chen, B. Li, L. Ye, D. Shou, Personal thermal management by thermally conductive composites: a review, Compos. Commun. 23 (2021), 100595, https://doi.org/10.1016/J.COCO.2020.100595.

[9] E. Delzendeh, S. Wu, A. Lee, Y. Zhou, The impact of occupants' behaviours on building energy analysis: a research review, Renew. Sustain. Energy Rev. 80 (2017) 1061–1071, https://doi.org/10.1016/J.RSER.2017.05.264.

[10] L. Lou, D. Shou, H. Park, D. Zhao, Y.S. Wu, X. Hui, R. Yang, E.C. Kan, J. Fan, Thermoelectric air conditioning undergarment for personal thermal management and HVAC energy saving, Energy Build. 226 (2020), 110374, https://doi.org/10.1016/J.ENBUILD.2020.110374.

[11] M.V. Bavaresco, S. D'Oca, E. Ghisi, R. Lamberts, Technological innovations to assess and include the human dimension in the building-performance loop: a review, Energy Build. 202 (2019), 109365, https://doi.org/10.1016/J.ENBUILD.2019.109365.

[12] V.L. Castaldo, I. Pigliautile, F. Rosso, F. Cotana, F. De Giorgio, A.L. Pisello, How subjective and non-physical parameters affect occupants' environmental comfort perception, Energy Build. 178 (2018) 107–129, https://doi.org/10.1016/J.ENBUILD.2018.08.020.

[13] M. Franke, C. Nadler, Towards a holistic approach for assessing the impact of IEQ on satisfaction, health, and productivity, Build. Res. Inf. (2020) 1–28, https://doi.org/10.1080/09613218.2020.1788917, 0.

[14] W. O'Brien, A. Wagner, M. Schweiker, A. Mahdavi, J. Day, M.B. Kjærgaard, S. Carlucci, B. Dong, F. Tahmasebi, D. Yan, T. Hong, H.B. Gunay, Z. Nagy, C. Miller, C. Berger, Introducing IEA EBC annex 79: key challenges and opportunities in the field of occupant-centric building design and operation, Build. Environ. 178 (2020), https://doi.org/10.1016/J.BUILDENV.2020.106738.

[15] W.S. Suen, G. Huang, Z. Kang, Y. Gu, J. Fan, D. Shou, Development of wearable air-conditioned mask for personal thermal management, Build. Environ. 205 (2021), 108236, https://doi.org/10.1016/J.BUILDENV.2021.108236.

[16] M. Schweiker, E. Ampatzi, M.S. Andargie, R.K. Andersen, E. Azar, V.M. Barthelmes, C. Berger, L. Bourikas, S. Carlucci, G. Chinazzo, L.P. Edappilly, M. Favero, S. Gauthier, A. Jamrozik, M. Kane, A. Mahdavi, C. Piselli, A.L. Pisello, A. Roetzel, A. Rysanek, K. Sharma, S. Zhang, Review of multi-domain approaches to indoor environmental perception and behaviour, Build. Environ. (2020), https://doi.org/10.1016/j.buildenv.2020.106804.

[17] M. Schweiker, G.M. Huebner, B.R.M. Kingma, R. Kramer, H. Pallubinsky, Drivers of diversity in human thermal perception – a review for holistic comfort models, Temperature 5 (2018) 308–342, https://doi.org/10.1080/23328940.2018.1534490.

[18] Y. Geng, W. Ji, Z. Wang, B. Lin, Y. Zhu, A review of operating performance in green buildings: energy use, indoor environmental quality and occupant satisfaction, Energy Build. 183 (2019) 500–514, https://doi.org/10.1016/J.ENBUILD.2018.11.017.

[19] S. Liu, S. Schiavon, H.P. Das, M. Jin, C.J. Spanos, Personal thermal comfort models with wearable sensors, Build. Environ. (2019), https://doi.org/10.1016/j.buildenv.2019.106281.

[20] T. Chaudhuri, Y.C. Soh, H. Li, L. Xie, Machine learning driven personal comfort prediction by wearable sensing of pulse rate and skin temperature, Build. Environ. 170 (2020), 106615, https://doi.org/10.1016/J.BUILDENV.2019.106615.

[21] J. Kim, Y. Zhou, S. Schiavon, P. Raftery, G. Brager, Personal comfort models: predicting individuals' thermal preference using occupant heating and cooling behavior and machine learning, Build. Environ. 129 (2018) 96–106, https://doi.org/10.1016/J.BUILDENV.2017.12.011.

[22] S.A. Mansi, G. Barone, C. Forzano, I. Pigliautile, M. Ferrara, A.L. Pisello, M. Arnesano, Measuring human physiological indices for thermal comfort assessment through wearable devices: a review, Meas. J. Int. Meas. Confed. 183 (2021), 109872, https://doi.org/10.1016/j.measurement.2021.109872.

[23] P.A. Abhang, B.W. Gawali, S.C. Mehrotra, Technological basics of EEG recording and operation of apparatus, introd. To EEG- speech-based emot, Recognit (2016) 19–50, https://doi.org/10.1016/B978-0-12-804490-2.00002-6.

[24] U. Herwig, P. Satrapi, C. Schönfeldt-Lecuona, Using the international 10-20 EEG system for positioning of transcranial magnetic stimulation, Brain Topogr. 16 (2003) 95–99, https://doi.org/10.1023/B:BRAT.0000006333.93597.9D.

[25] K.J. Mathewson, A. Hashemi, B. Sheng, A.B. Sekuler, P.J. Bennett, L.A. Schmidt, Regional electroencephalogram (EEG) alpha power and asymmetry in older adults: a study of short-term test-retest reliability, Front. Aging Neurosci. 7 (2015) 177, https://doi.org/10.3389/FNAGI.2015.00177/BIBTEX.

[26] B. Lv, C. Su, L. Yang, T. Wu, Effects of stimulus mode and ambient temperature on cerebral responses to local thermal stimulation: an EEG study, Int. J. Psychophysiol. 113 (2017) 17–22, https://doi.org/10.1016/j.ijpsycho.2017.01.003.

[27] J.R. Lim, G.H. Baek, E.S. Jeon, Analysis of the correlation between thermal sensations and brain waves via EEG measurements, Int. J. Appl. Eng. Res. 13 (2018) 6069–6075. http://www.ripublication.com. (Accessed 1 April 2022).

[28] S.A. Mansi, I. Pigliautile, C. Porcaro, A.L. Pisello, M. Arnesano, Application of wearable EEG sensors for indoor thermal comfort measurements, Acta IMEKO 10 (2021) 214–220, https://doi.org/10.21014/acta_imeko.v10i4.1180.

[29] Y.J. Son, C. Chun, Research on electroencephalogram to measure thermal pleasure in thermal alliesthesia in temperature step-change environment, Indoor Air 28 (2018) 916–923, https://doi.org/10.1111/INA.12491.

[30] Research in physiology signal change of thermal-comfort evaluation by air conditioner temperature change, Korea Science, (n.d.), Journal of Fisheries and Marine Sciences Education (2022), https://www.koreascience.kr/article/JAKO200610103441292.page. (Accessed 1 April 2022).

[31] J.R. Lim, G.H. Baek, E.S. Jeon, Analysis of the correlation between thermal sensations and brain waves via EEG measurements, Int. J. Appl. Eng. Res. 13 (2018) 6069–6075.

[32] A. Gautam, ECG analysis using continuous wavelet Transform (CWT), IOSR J. Eng. (2012) 632–635, https://doi.org/10.9790/3021-0204632635, 02.

[33] B.J. Singstad, N. Azulay, A. Bjurstedt, S.S. Bjørndal, M.F. Drageseth, P. Engeset, K. Eriksen, M.Y. Gidey, E.O. Granum, M.G. Greaker, A. Grorud, S.O. Hewes, J. Hou, A.M. Llop Recha, C. Matre, A. Seputis, S.E. Sørensen, V. Thøgersen, V.M. Joten, C. Tronstad, Ø.G. Martinsen, Estimation of heart rate variability from finger photoplethysmography during rest, mild exercise and mild mental stress, J. Electr. Bioimpedance. 12 (2021) 89, https://doi.org/10.2478/JOEB-2021-0012.

[34] Z. Zhang, Z. Pi, B. Liu, TROIKA: a general framework for heart rate monitoring using wrist-type photoplethysmographic signals during intensive physical exercise, IEEE Trans. Biomed. Eng. 62 (2015) 522–531, https://doi.org/10.1109/TBME.2014.2359372.

[35] A.A. Alian, K.H. Shelley, Photoplethysmography, Best Pract. Res. Clin. Anaesthesiol. 28 (2014) 395–406, https://doi.org/10.1016/J.BPA.2014.08.006.

[36] L. Carlos, M. Vanderlei, C.M. Pastre, R.A. Hoshi, T. Dias De Carvalho, M. Fernandes De Godoy, M.V. Rua Bela, Basic notions of heart rate variability and its clinical applicability, Rev. Bras. Cir. Cardiovasc. 24 (2009) 205–217.

[37] Y. Yao, Z. Lian, W. Liu, Q. Shen, Experimental study on physiological responses and thermal comfort under various ambient temperatures, Physiol. Behav. 93 (2008) 310–321, https://doi.org/10.1016/J.PHYSBEH.2007.09.012.

[38] W. Liu, Z. Lian, Y. Liu, Heart rate variability at different thermal comfort levels, Eur. J. Appl. Physiol. 103 (2008) 361–366, https://doi.org/10.1007/S00421-008-0718-6.

[39] Y. Yao, Z. Lian, W. Liu, C. Jiang, Y. Liu, H. Lu, Heart rate variation and electroencephalograph - the potential physiological factors for thermal comfort study, Indoor Air 19 (2009) 93–101, https://doi.org/10.1111/J.1600-0668.2008.00565.X.

[40] H. Zhu, H. Wang, Z. Liu, D. Li, G. Kou, C. Li, Experimental study on the human thermal comfort based on the heart rate variability (HRV) analysis under different environments, Sci. Total Environ. (2018), https://doi.org/10.1016/j.scitotenv.2017.10.208.

[41] G. Turpin, T. Grandfield, Electrodermal activity, Encycl. Stress (2007) 899–902, https://doi.org/10.1016/B978-012373947-6.00139-2.

[42] J.H. Choi, D. Yeom, Study of data-driven thermal sensation prediction model as a function of local body skin temperatures in a built environment, Build. Environ. 121 (2017) 130–147, https://doi.org/10.1016/J.BUILDENV.2017.05.004.

[43] Office thermal comfort state analysis based on physiological signal from wearable device – Fingerprint, Yonsei University, (n.d.), https://yonsei.pure.elsevier.com/en/publications/office-thermal-comfort-state-analysis-based-on-physiological-sign/fingerprints/?sortBy=alphabetically. (Accessed 1 April 2022).

[44] B. Kobas, S.C. Koth, K. Nkurikiyeyezu, G. Giannakakis, T. Auer, Y. Yamamoto, Effect of exposure time on thermal behaviour: a psychophysiological approach, 2, 2021, Signals 2 (2021) 863–885, https://doi.org/10.3390/SIGNALS2040050, 863–885.

[45] Q. Xu, L. Chen, H. Chen, B.J. Dewancker, Exercise thermal sensation: physiological response to dynamic-static steps at moderate exercise, Int. J. Environ. Res. Publ. Health 18 (2021), https://doi.org/10.3390/IJERPH18084239.

[46] O.E. Krigolson, C.C. Williams, A. Norton, C.D. Hassall, F.L. Colino, Choosing MUSE: validation of a low-cost, portable EEG system for ERP research, Front. Neurosci. 11 (2017), https://doi.org/10.3389/FNINS.2017.00109.

[47] M. Garbarino, M. Lai, S. Tognetti, R. Picard, D. Bender, Empatica E3 - a wearable wireless multi-sensor device for real-time computerized biofeedback and data acquisition. https://doi.org/10.4108/icst.mobihealth.2014.257418, 2014, 39-42.

[48] E.E. van der Wall, The brain-heart connection; a round trip, Undefined 19 (2011) 269–270, https://doi.org/10.1007/S12471-011-0161-X.

[49] K. Ritz, M.A. van Buchem, M.J. Daemen, The heart-brain connection: mechanistic insights and models, Neth. Heart J. 21 (2013) 55–57, https://doi.org/10.1007/S12471-012-0348-9.

[50] Chapter 01: heart-brain communication, HeartMath Institute, (n.d.), https://www.heartmath.org/research/science-of-the-heart/heart-brain-communication/. (Accessed 17 January 2022).
[51] R. Paus, Exploring the "brain-skin connection": leads and lessons from the hair follicle, Curr. Res. Transl. Med. 64 (2016) 207–214, https://doi.org/10.1016/J.RETRAM.2016.10.003.
[52] D. Roosterman, T. Goerge, S.W. Schneider, N.W. Bunnett, M. Steinhoff, Neuronal control of skin function: the skin as a neuroimmunoendocrine organ, Physiol. Rev. 86 (2006) 1309–1379, https://doi.org/10.1152/PHYSREV.00026.2005/ASSET/IMAGES/LARGE/Z9J0040624160008. JPEG.
[53] A.A. Romanovsky, The thermoregulation system and how it works, Handb. Clin. Neurol. 156 (2018) 3–43, https://doi.org/10.1016/B978-0-444-63912-7.00001-1.
[54] A.L. Pisello, I. Pigliautile, M. Andargie, C. Berger, P.M. Bluyssen, S. Carlucci, G. Chinazzo, Z. Deme Belafi, B. Dong, M. Favero, A. Ghahramani, G. Havenith, A. Heydarian, D. Kastner, M. Kong, D. Licina, Y. Liu, A. Luna-Navarro, A. Mahdavi, A. Nocente, M. Schweiker, M. Touchie, M. Vellei, F. Vittori, A. Wagner, A. Wang, S. Wei, Test rooms to study human comfort in buildings: a review of controlled experiments and facilities, Renew. Sustain. Energy Rev. 149 (2021), 111359, https://doi.org/10.1016/J.RSER.2021.111359.
[55] F. Vittori, C. Chiatti, I. Pigliautile, A.L. Pisello, The NEXT.ROOM: design principles and systems trials of a novel test room aimed at deepening our knowledge on human comfort, Build. Environ. 211 (2022), 108744, https://doi.org/10.1016/J.BUILDENV.2021.108744.
[56] ISO - ISO 7730, Ergonomics of the Thermal Environment — Analytical Determination and Interpretation of Thermal Comfort Using Calculation of the PMV and PPD Indices and Local Thermal Comfort Criteria, 2005 (n.d.), https://www.iso.org/standard/39155.html. (Accessed 24 February 2022).
[57] E. Ratti, S. Waninger, C. Berka, G. Ruffini, A. Verma, Comparison of medical and consumer wireless EEG systems for use in clinical trials, Front. Hum. Neurosci. 11 (2017), https://doi.org/10.3389/FNHUM.2017.00398.
[58] O.E. Krigolson, C.C. Williams, A. Norton, C.D. Hassall, F.L. Colino, Choosing MUSE: validation of a low-cost, portable EEG system for ERP research, Front. InNeuroscience. 11 (2017), https://doi.org/10.3389/fnins.2017.00109.
[59] (n.d.), https://mind-monitor.com/.
[60] F. Faul, E. Erdfelder, A.G. Lang, A. Buchner, G*Power 3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences, Behav. Res. Methods 39 (2007) 175–191, https://doi.org/10.3758/BF03193146.
[61] EEG-Signal Processing Lecture Notes for BSP, Chapter 5 Master Program Data Engineering, (n.d.).
[62] hrv-analysis · PyPI (n.d.), https://pypi.org/project/hrv-analysis/. (Accessed 29 October 2021).
[63] M. Malik, A. John Camm, J. Thomas Bigger, G. Breithardt, S. Cerutti, R.J. Cohen, P. Coumel, E.L. Fallen, H.L. Kennedy, R.E. Kleiger, F. Lombardi, A. Malliani, A. J. Moss, J.N. Rottman, G. Schmidt, P.J. Schwartz, D.H. Singer, Heart rate variability: standards of measurement, physiological interpretation, and clinical use, Circulation 93 (1996) 1043–1065, https://doi.org/10.1161/01.cir.93.5.1043.
[64] J. Sztajzel, Heart rate variability: a noninvasive electrocardiographic method to measure the autonomic nervous system, Swiss Med. Wkly. 134 (2004) 514–522, https://doi.org/2004/35/smw-10321.
[65] M. Malik, A. John Camm, J. Thomas Bigger, G. Breithardt, S. Cerutti, R.J. Cohen, P. Coumel, E.L. Fallen, H.L. Kennedy, R.E. Kleiger, F. Lombardi, A. Malliani, A. J. Moss, J.N. Rottman, G. Schmidt, P.J. Schwartz, D.H. Singer, Heart rate variability: standards of measurement, physiological interpretation, and clinical use, Circulation 93 (1996) 1043–1065, https://doi.org/10.1161/01.CIR.93.5.1043.
[66] S. Taylor, N. Jaques, W. Chen, S. Fedor, A. Sano, R. Picard, Automatic identification of artifacts in electrodermal activity data, 2015-Novem, in: Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. EMBS., 2015, pp. 1934–1937, https://doi.org/10.1109/EMBC.2015.7318762.
[67] A. Greco, G. Valenza, A. Lanata, E.P. Scilingo, L. Citi, CvxEDA: a convex optimization approach to electrodermal activity processing, IEEE Trans. Biomed. Eng. 63 (2016) 797–804, https://doi.org/10.1109/TBME.2015.2474131.
[68] ISO, Evaluation of Measurement Data — Guide to the Expression of Uncertainty in Measurement, vol. 50, Int. Organ. Stand, Geneva, 2008, p. 134. http://www.bipm.org/en/publications/guides/gum.html.
[69] M. Brzezinski, The Chen-Shapiro test for normality, STATA J. 12 (2012) 368–374, https://doi.org/10.1177/1536867X1201200302.
[70] H. Arsham, M. Lovric, Bartlett's test, Int. Encycl. Stat. Sci. (2011) 87–88, https://doi.org/10.1007/978-3-642-04898-2_132.
[71] P.A. Games, Alternative analyses of repeated-measure designs by ANOVA and MANOVA, Stat. Methods Longitud. Res. (1990) 81–121, https://doi.org/10.1016/B978-0-12-724960-5.50008-1.
[72] E. Ostertagová, O. Ostertag, J. Kováč, Methodology and application of the Kruskal-Wallis test, Appl. Mech. Mater. 611 (2014) 115–120, https://doi.org/10.4028/WWW.SCIENTIFIC.NET/AMM.611.115.
[73] D.E. Critchlow, M.A. Fligner, On distribution-free multiple comparisons in the one-way analysis of variance. https://doi.org/10.1080/03610929108830487, 2007, 20,127-139.
[74] W. von Rosenberg, T. Chanwimalueang, T. Adjei, U. Jaffer, V. Goverdovsky, D. P. Mandic, Resolving ambiguities in the LF/HF ratio: LF-HF scatter plots for the categorization of mental and physical stress from HRV, Front. Physiol. 8 (2017), https://doi.org/10.3389/FPHYS.2017.00360/FULL.